Status	Finished
Started	Tuesday, 20 October 2026, 9:40 PM
Completed	Tuesday, 20 October 2026, 9:42 PM
Duration	2 mins 30 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O(1) ✓
d. O🥱

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓
b. The list is sorted
c. The search continues in the left sublist
d. The search continues in the right sublist

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues in the left or right sublist ✓
b. The list is sorted
c. The search stops
d. The search continues from the beginning

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search works on unsorted lists
- b. Binary search can find multiple instances of the target element
- c. Binary search does not require dividing the list
- d. Binary search has a lower time complexity on large, sorted lists ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and sorted
- b. When the list is small and unsorted ✓
- c. When the list is large and unsorted
- d. When the list is large and sorted

Question **6**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O👎
- c. O(n log n)
- d. O(1)

Question **7**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. log n
- b. n-1
- c. n/2
- d. n ✓

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is not the array at all

Question **9**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the search algorithm finishes

b. When the list is sorted

c. When the element is found in the list ✓

d. When the list contains duplicate elements

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓

b. 500

c. 1000

d. 100

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining

b. Finding

c. Searching ✓

d. Discovering

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search requires the list to be sorted.
d. Linear search divides the list into halves.

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Bubble search
c. Linear search ✓
d. Quick search

Question **14**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the middle element in the list ✓
c. Sort the list
d. Compare the target element with the first element in the list

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. n/2
c. 1
d. log n